

Detoxification

First stage

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FUNCTIONS OF THE LIVER

Excretory
Function

Synthetic
Function

Metabolic
Function

Detoxification
Function

Storage
Function

Detoxification

Detoxification: is the processes whereby the toxic substances are rendered less harmful, more water soluble, and more easily excreted in urine.

Toxicants that are lipophilic, non-polar, and of low molecular weight are readily absorbed through the cell membranes of the skin, gastrointestinal tract, and lung. Lipophilic toxicants are hard for the body to eliminate and can accumulate to hazardous levels.

In general, biotransformation reactions generate more polar metabolites, that are readily excreted from the body.

The liver plays the most important role. Biotransformation is thus a key body defense mechanism.

Xenobiotics

Xenobiotics are compounds which may be accidentally ingested or taken as drugs or compounds produced in the body by bacterial mechanism. (Greek, xenos = strange).

Biotransformation

Is not synonymous with detoxification, since some of the reactions involved may be led to activation or even toxication.

An example is the;
biotransformation of vinyl chloride to
vinyl chloride epoxide, which
covalently binds to DNA and RNA, a
step leading to cancer of the liver.

The compounds that are detoxified include:

- A. Compounds accidentally ingested like preservatives, cosmetics, pollutants and food additives.
- B. Drugs which are taken for therapeutic purposes.**
- C. Compounds produced in the body which are be eliminated, e.g. bilirubin, free radicals and steroids.

**D- Compounds produced by bacterial metabolism,
e.g. amines produced by decarboxylation of
amino acids:**

Histidine to histamine

Lysine to cadaverine

Tyrosine to tyramine

Phenylalanine to phenol

Excess toxins get stored in;

- Liver, joints, blood, tissue, muscle, fat cells, bone marrow, and central nervous system.

Usually, accumulation of toxins (toxin burden) depends on;

☐Genetics

☐Age

☐Environment

☐Lifestyle such as; diet, exercise, stress....ect

Side effects and signs of toxicity:

A- Symptoms of poor body detoxification (failure to detoxify symptoms)

- Insomnia, difficulty getting to sleep at night
- Skin problems: acne, rashes
- Constipation
- Excess body fat, fatty liver, gallstones

- Depression, anxiety
- Fatigue
- Allergic reactions
- Mental confusion
- poor concentration & memory

B- Failure to detoxify conditions (lot of toxicity can):

- Inflammatory conditions
- Rheumatoid arthritis
- Parkinson's disease
- Alzheimer's disease

- Cancer
- Abnormal metabolism
- Body can not absorb the full nutrition
- **Immune system decrease and unbalanced**

The transformation of a specific xenobiotic can be either beneficial or harmful, and perhaps both depending on the acetaminophen, a commonly used drug to reduce pain and fever. When the prescribed doses are taken, the desired therapeutic response is observed with little or no toxicity.

However, when excessive doses of acetaminophen are taken, hepatotoxicity can occur. This is because acetaminophen normally undergoes rapid biotransformation with the metabolites quickly eliminated in the urine and feces.

Enzyme systems

The cytochrome P450 (CYP 450) enzyme family is involved in the biotransformation reactions. They are heme-containing membrane proteins, localized in the smooth endoplasmic reticulum of liver.

CYT P-450



The reaction catalyzed by P-450 flavoprotein.

At least 20 gene families for P-450 are identified, most isoforms are inducible. Phenobarbital causes hypertrophy of endoplasmic reticulum, and increased activity of P-450. some of the isoforms of the enzyme exhibit low catalytic activity. This explains the variation in drug responses among patients.

Phases of detoxification processes.

Phase I

Reactions include

oxidation,

reduction,

hydrolysis

and

hydroxylation.

Phase II

Includes conjugation reactions involving conjugating molecules such as glucuronic acid, amino acids, sulphate, etc....

Generally, detoxification of a compound involves phase I, phase II reactions, followed by;

Phase III elimination of toxins.

A brief consideration of the various mechanisms are given below;

FAT-SOLUBLE TOXINS

Phase 1 (Cytochrome P450 Enzymes)

Oxidation
Reduction
Hydrolysis
Hydration
Dehalogenation

Nutrients Needed

- Vitamins B2, B3, B6, B12
- Folic Acid
- Glutathione
- Flavonoids

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Phase 2

(Conjugation Pathways)

Sulfation
Glucoronidation
Glutathione Conjugation
Acetylation
Amino Acid Conjugation
Methylation

Nutrients Needed

- | | | |
|---------------|-------------------|--------------|
| • Methionine | • Vitamin B5, B12 | • Glutamine |
| • Cysteine | • Vitamin C | • Folic Acid |
| • Magnesium | • Glycine | • Choline |
| • Glutathione | • Taurine | |

WATER-SOLUBLE WASTE

Eliminated via:

Urine
Bile
Stool

1- Oxidation

Are large number of foreign compounds are detoxified by oxidation. These include;

- Alcohols
- Aldehydes
- Amines
- Aromatic hydrocarbons

Ethanol may be completely oxidized in the body to CO_2 , H_2O , and energy.

Aromatic alcohols, and aldehydes are oxidized to the corresponding acids.

Aliphatic amines are completely oxidized in the body. This is carried out by the enzyme, mono amino oxidase (MAO).

Aromatic hydrocarbons such as benzene is oxidized to muconic acid which involves opening of the ring.

Benzoic acid may be conjugated with amino acid (glycine) to form “hippuric acid”, and excreted in urine.

2- Reduction

Is less common and two important examples for reduction are;

Picric acid $\longrightarrow$ picramic acid

Chloral $\longrightarrow$ trichloromethyl alcohol

Note; certain of the reduced metabolites instead of being less toxic, may be more toxic.

3-Hydrolysis

These are quite a number of therapeutic compounds, used as drugs, which undergo hydrolysis, usually in liver.

Acetyl salicylic acid $\longrightarrow$ salicylic acid + acetic acid
(*Aspirin*)

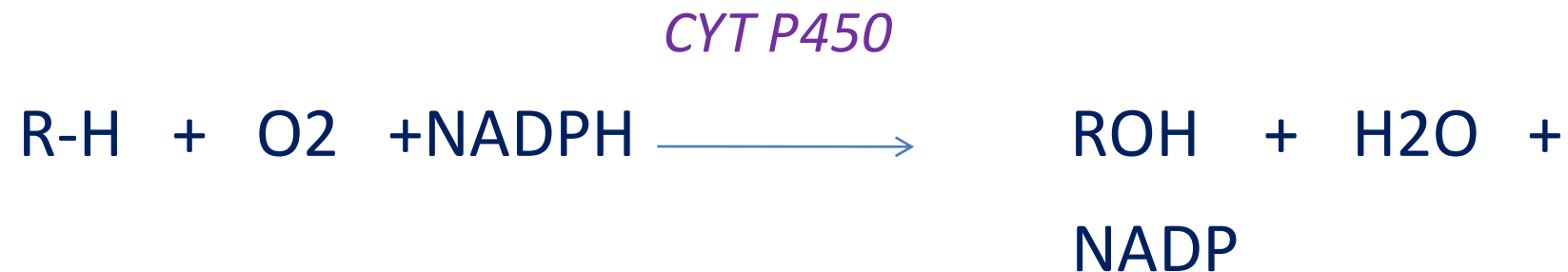
• **Atropine** $\longrightarrow$ **tropic acid + tropin**
(troyl troyate)

• **Digitalis** $\longrightarrow$ **sugar + a glycone**
(a cardiac glycoside) *(non sugar component)*

• **Procine** $\longrightarrow$ **P-amino benzoic acid +
diethylamino ethanol**

4- Hydroxylation

Is catalyzed by mixed function oxidase (mono-oxygenase) system present in the microsomes of the hepatic cells. Cytochrome P450 (CYT P 450) are heme proteins and is an important component of this system.



- Cytochrome P 450 (CYT P450) is an inducible enzyme.

Its synthesis is increased by the administration of drugs such as phenobarbital. Several drugs are detoxified by the mono –oxygenase system.

For example;

CYT P450

Meprobamate $\longrightarrow$ hydroxy meprobamate

Meprobamate; a tranquilizer used in psychiatric disorders is excreted largely as the oxidation product hydroxy meprobamate.

CYT P450



•***Note;***

Drugs such as phenobarbital and chloranthrene are detoxified by hydroxylation.

Characteristic features of Cytochrome P 450 (CYT P450) system

- Present in endoplasmic reticulum (ER), microsomal fraction of liver (highest concentration).
- Chemically they are “haemoproteins”
- Enzyme is “NADPH-cyt P 450 reductase”.
- The enzyme requires NADPH.
- It is an inducible enzyme.

Biochemical importance of mono-oxygenase

Cyt-P 450 system

Recently another cytochrome P450 species has been found called as cytochrome P 448. this species has been found to be specific for metabolism of polycyclic aromatic hydrocarbons (PAHs). Hence this species has been named as aromatic hydrocarbon hydroxylase (AHH).

Clinical importance

It has been found to be very important enzyme for metabolism poly cyclic aromatic hydrocarbons (PAHs) and in carcinogenesis produced by these agents.

Studies have shown that:

- In the lungs of cigarette smoking the enzyme may be involved in conversion of inactive (PAHs) (procarcinogens) present in cigarette smoke to active carcinogens by hydroxylation reactions. Smokers were found to have higher levels of this enzyme in cells and tissues than non-smokers.

- Some reports have shown that activity of this enzyme is increased (induced) in placenta of pregnant women who are cigarette smokers and thus fetus is exposed to potentially harmful metabolites (carcinogens).

Phase II reactions: Conjugation

Is a process by which the foreign molecules or its metabolites are coupled with a conjugating agent and converted to soluble, nontoxic derivatives which are easily excreted in urine.

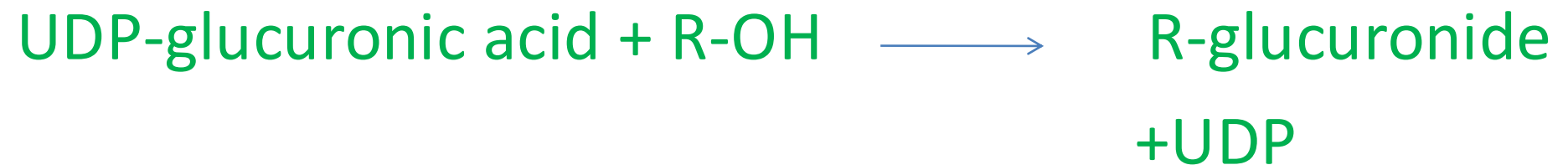
Features of conjugation:

- Various conjugating agents are available in the body and some of them are synthesized in the body, e.g. D-glucuronic acid formed from glucose by uronic acid pathway. Certain amino acids as glycine, cysteine, can be available from dietary proteins/ or breakdown of tissue proteins or synthesized.

- Conjugation reaction principally occurs in liver and to some extent it can occur in kidneys also.
- Conjugation produces less toxic, more soluble compounds which are excreted.
- Conjugation can occur independently or it can follow oxidation, reduction or hydroxylation of a compound.

1- Glucuronic acid:

Glucuronide conjugation is the most common phase II reactions. Bilirubin is a good example for a compound conjugated and excreted as its glucuronide. Glucuronic acid can conjugate with hydroxyls (both phenolic and alcoholic), carbonyl, sulfhydryl and amino compounds. The glucuronic acid is added to xenobiotics by UDP-glucuronyl transferases, present in the endoplasmic reticulum.



2- Sulfate:

Sulfate conjugation occurs with many xenobiotic. In general, **sulfation** decreases the toxicity of xenobiotic. Unlike **glucuronic acid** conjugates that are often eliminated in the bile, the highly polar sulfate conjugates are readily secreted in the urine.

Active sulfate acts as the donor, chemically it is 3-phospho adenosine-5-phosphosulfate (PAPS)

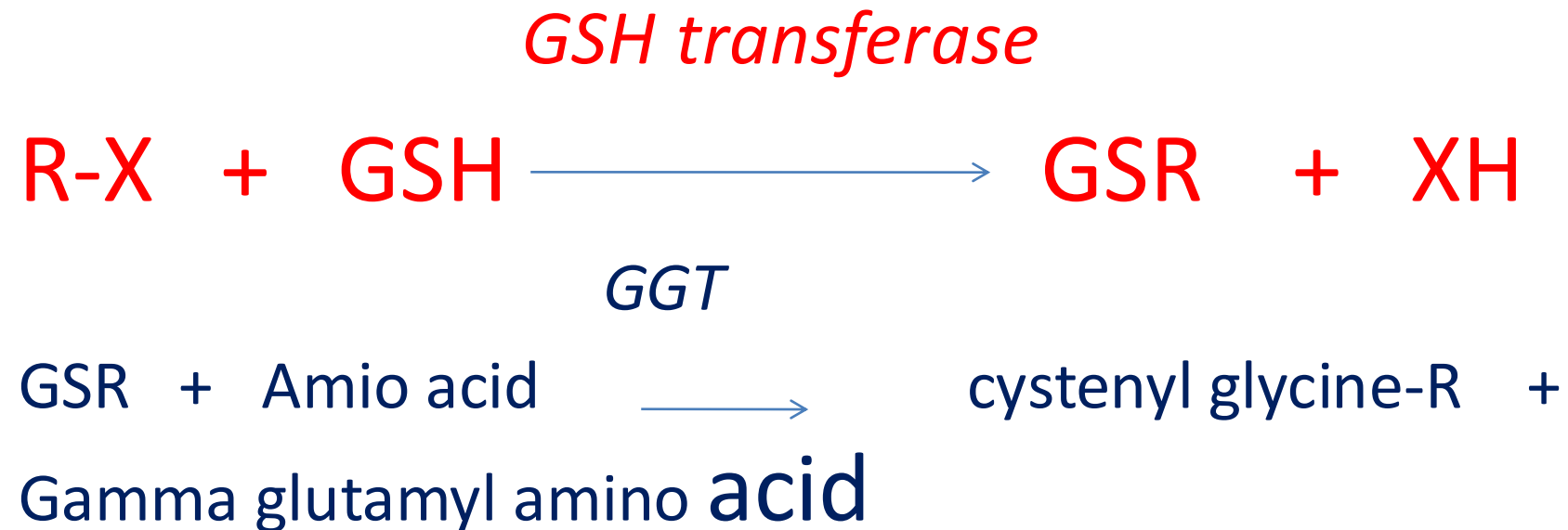
Phenol $\longrightarrow$ Phenyl sulfuric acid

Indole $\longrightarrow$ Indoxyl $\longrightarrow$ Indoxyl sulfuric acid

3- Cysteine and glutathione:

The cysteine is derived from glutathione (GSH), which is the active conjugating agent.

For example;



GGT: Gamma glutamyl transferase

Note;

If the potentially toxic xenobiotic are not conjugated with GSH, they would be free to combine covalently with DNA, RNA or cell proteins and can produce serious cell damage.

Glutathione (GSH): is thus an important defense mechanism against certain toxic compounds, such as some drugs and carcinogens.

4- Acetylation:

Conjugation with acetyl group is taking place with drugs like sulfanilamide, and isoniazid.

5- Conjugation with glycine:

Benzoic acid is conjugated with glycine to form hippuric acid (benzoyl glycine, which is excreted with urine.

6- Conjugation with glutamine:

Phenyl acetic acid is conjugated with glutamine to form phenyl acetyl glutamine, which is excreted in urine.

7- Methylation reactions:

- Methylation as a detoxication process through limited in the body, at the same time is quite important. Usual methyl donor is “S-adenosyl methionine” (“active” methionine).
- Methylation of heterocyclic N-atom of compounds of the pyrimidine and quinoline types e.g.,nicotinamide. This occurs also with other heterocyclic aromatic compounds e.g., histamine.

- O-methylation of certain naturally occurring amines (with phenolic hydroxyl group), e.g., epinephrine and nor-epinephrine and their metabolites are methylated at the phenolic hydroxyl group.
- O-methylation of natural estrogens.

Phase III: Elimination of toxins

Once conjugated toxins must be eliminated from the body via the bile (colon) or kidneys, or skin.

Adequate bile flow is important for detoxification, dietary fiber bind toxins in colon and water to flush toxins from the system.

Detoxication of heavy metals:

- Selenium poisoning: selenium poisoning develops due to high feeding of products obtained from the soil having high contents of selenium.
- Reason for toxicity: selenium replaces Sulphur in cysteine and methionine in body tissues.

- Dithio propanol: also known as;
2,3-mercaptopropanol or (British-anti-lewesite “BAL”)

“BAL” was used as a detoxicant for certain war poisons (chemical warfare). It is now known that “BAL” is valuable for removal of several toxic heavy metals, such as; arsenic (As), gold (Au), mercury (Hg), cadmium (Cd).

Drug metabolism

Most drugs, and particularly oral drugs, are modified or degraded in the liver. Medications that are metabolized by the liver must be used with caution in patients with hepatic disease; such patients may need lower doses of the drug, because it can cause potential toxication.

The metabolism of paracetamol is an example of potential toxication. Paracetamol is metabolized primarily in the liver, via phase II metabolism, where its major metabolites include sulphate and glucuronide conjugates, which are excreted by the kidneys.

A small yet significant amount is metabolized via the hepatic CYP enzyme system, to a minor alkylating metabolic N-acetyl-P-benzo-quinone imine (NAPQI). Normally NAPQI is detoxified by conjugation with glutathione.

In cases of paracetamol toxicity (overdose);

The sulphate and glucuronide pathways become saturated, and more paracetamol is shunted to the CYP system to produce NAPQI. As a result, hepatocellular supplies of glutathione become exhausted and NAPQI is free to react with cellular membrane molecules, resulting in widespread hepatocyte damage and death, and leading to acute hepatic necrosis.

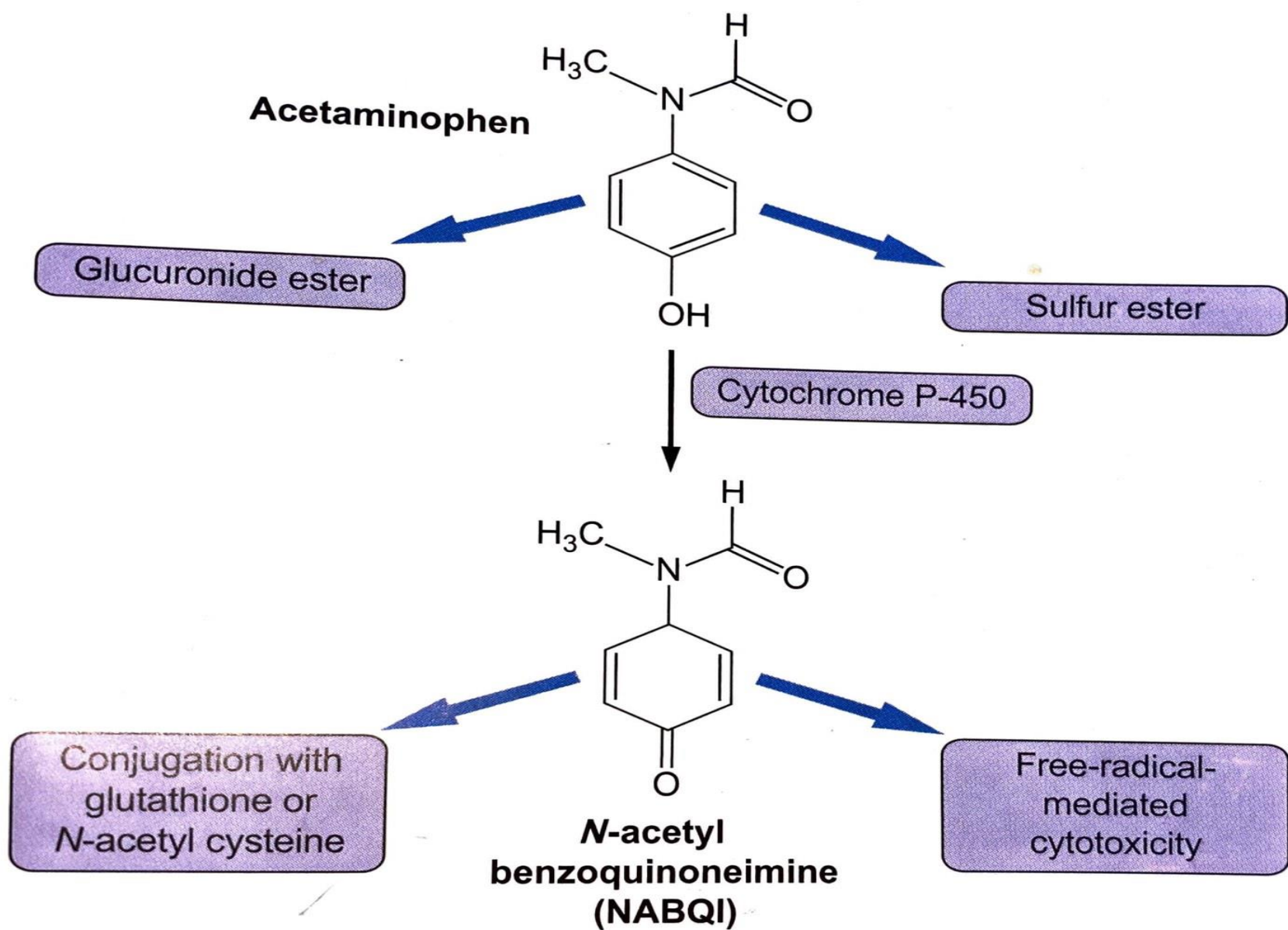


Fig. 30.7 Metabolism of acetaminophen (paracetamol).

What can we do to reduce toxicity?

- Reduce take toxicity
- Eat less; smoked, carcinogens nitrite, preservatives, and processed food.
- Eat more; fresh fruits and vegetables
- Contact less with toxicity
- Drink fresh

Detoxes and let our inside body has a clean environment for good mode and comfortable

With your health and wellness